

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	i		% O ₂ saturation stays above 95% at altitudes/ O ₂ affinity stays high up to 1500m/ Enables humans to {live/ survive} at altitude		1		1		
		ii		Increased red blood cell count/ more haemoglobin/ haemoglobin has an increased affinity for oxygen		1		1		
		iii		(gut lumen) highly anaerobic/ low concentration of oxygen (in gut) (1) higher affinity (for O ₂ than humans)/ can absorb any available O ₂ / higher saturation at lower partial pressures (1)		2				
	(b)	i		{Structures / molecules} with no close phylogenetic links / have evolved from different origins/ different structures/ ref to analogous structures(1) adapted to carry out {same/ similar} function (1)	2			2		
		ii		As temp rises {higher ppO ₂ / higher concentration of oxygen} (needed {to fully saturate/ reach saturation/ reach 96%}).		1		2		
		iii		As temperature rises more O ₂ released (to tissues).		1		1		
				Question 5 total	2	6	0	8	0	0